

Maria Monica Manlises

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EDUCATION

De La Salle University (DLSU), Manila

Expected Feb. 2027

Bachelor of Science (Honors) in Computer Science and Master of Science in Computer Science

- CGPA: 3.968/4.0 (Summa Cum Laude Standing)
- Archer Achiever Scholar (Given to the top 100 applicants upon university entry)

Universitas Gadjah Mada, Indonesia

July 2024 - Aug. 2024

Department of Computer Science and Electronics (DCSE) Summer Course on AI and Robotics 2024

- 120-hour online course on AI for Smart City Development to Support Sustainable Cities and Communities

RESEARCH EXPERIENCE

DLSU Center for Language Technologies

Sept. 2024 - present

Student Researcher under Dr. Ethel Ong

Constrained Phonics Story Generation and Evaluation using Large Language Models

- Generated 39,096 stories using six LLMs with 16 zero-shot prompts focusing on 112 grapheme-phoneme pairs
- Evaluated model performance using 12 metrics focused on task relevance, age-appropriateness, readability, and narrative quality

Mohamed Bin Zayed University for Artificial Intelligence (MBZUAI)

June 2025 - July 2025

Undergraduate Research Intern Under Prof. Thamar Solorio, Dr. Injy Hamed

Evaluating Code-Switching in LLMs: From Benchmarks to Cross-Lingual Alignment

- Created 400 multilingual code-switched QA pairs from Global-MMLU-Lite across five languages (L1) – Arabic, French, Hindi, Portuguese, and Filipino – evaluating fluency, content preservation, and generation type
- Evaluated four LLMs on answering English, L1, and code-switched questions, showing best performance on the English dataset and worst performance on the L1 dataset across all LLMs
- Tested the impact of human-generated code-switching and cross-lingual alignment, finding little effect on accuracy with variation by script and language resources

DLSU Dr. Andrew L. Tan Data Science Institute (ALTDSI)

May 2023 - Sep. 2024

Research Assistant under Dr. Macario Cordel II

A Prediction Model-based Decision Tool for Airline Fuel Consumption Management

- Performed exploratory data analysis and pre-processing on 70,000 flight data from a partner airline
- Designed and executed MySQL database models for 15 aircraft types across 4 levels of granularity
- Designed prediction models, with the single-step autoregressive model producing the highest R^2 value

PUBLICATIONS

- Manlises, M., Ong, E. (2026). Zero shot phonics: Evaluating constraint-adherent phonics story generation in large language models. *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2026)*, USA. Association for Computational Linguistics. Accepted.
- Lasala, K., Go, D., Manlises, M., Uy, R. (2026). Parallel permutation testing of one-way ANOVA using CUDA. *Proceedings of the 26th Philippine Computing Science Congress, Philippines, Philippines*. Computing Society of the Philippines. In press.
- Abello, H., Asturiano, C., Calamiong, Y., Cheng, S., Cruz, A., De Leon, F., Dichoso, A., Encinas, R., Ermitaño, K., Espiritu, J., Estrera, D., Go, D., Juachon, J., Lasala, K., Lejano, E., **Manlises, M.**, Mantuano, J., Ong, C., Pineda, R., ... Oco, N. (2025). Rule-based pattern matching in an automata and formal languages course: Applications towards addressing the Sustainable Development Goals. *Proceeding of 2025 IEEE International Symposium on Technology and Society (ISTAS)*, USA, 1-6. Institute of Electrical and Electronics Engineers. <https://doi.org/10.1109/ISTAS65609.2025.11269652>
- **Manlises, M.**, Gonzales, M., & Lim, L. (2025). DLSU at BEA 2025 shared task: Towards establishing baseline models for pedagogical response evaluation tasks. *Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025)*, Austria, 1260-1265. Association for Computational Linguistics. <https://aclanthology.org/2025.bea-1.101/>
- Dy, S., Encinas, R., Go, D., Lasala, K., **Manlises, M.**, & Deja, J. (2024). Point n Move: Designing a glove-based pointing device. *Proceedings of CHIRP 2024: Transforming HCI Research in the Philippines Workshop, Philippines*, 49-58. CHI MNL. <https://doi.org/10.48550/arXiv.2412.00501>

PROJECTS

Filipino Tokenization and Embedding Analysis

Investigated limitations of Gemma-3-1B-IT tokenization for Filipino morphology, using linguistic analysis and PCA/t-SNE visualization to motivate low-resource-aware modelling.

Converting English Words to Phonemic Pronunciation

Developed a seq2seq model for English-to-phoneme conversion, comparing RNN, GRU, and LSTM. LSTM outperformed other models with 54.09% sequence accuracy, 82.17% character accuracy, a 2.07 edit distance, and an 86.40% BLEU score.

COMPETITIONS

- *Best Team Award* - Undergraduate Research Internship Program 2025 (MBZUAI)
- *Most Popular Project Pitch* - Asian Undergraduate Symposium 2024 (NUS)
- *1st Runner Up* - P&G NEXT Leadership Camp 2024 (*ERD construction, data population, PowerBI analysis*)
- *Top 25* - 2022 Shopee Ultimate Case Challenge (*ETA prediction improvement through machine learning*)
- *Bronze Award* - 2018 Bulgaria International Mathematics Competition Individual Contest
- *Champion* - 2018 Bulgaria International Mathematics Competition Team Contest

SKILLS

- **Languages:** Python, C, C++, Java, Go, SQL, Arduino, Prolog, Verilog, x86-64, HTML, CSS, Javascript, MERN, PERL
- **Libraries:** Pandas, Scikit-learn, NLTK, PyTorch, TensorFlow, Keras, SciPy, Matplotlib
- **Software Tools:** BigQuery, LookerStudio, Figma, Canva, Adobe Creative Cloud, Notion, Jira, Microsoft Office, Google Suite

INDUSTRY EXPERIENCE

Foodpanda (Delivery Hero Philippines, Inc.) April 2024 - June 2024

Data Analytics Intern – Logistics and Operations

- Conducted data extractions, cleaning, and visualizations using BigQuery and LookerStudio
- Automated a key operations tracker cutting weekly monitoring time by 90% and reducing manual data processing
- Provided data-driven insights to support decision-making for the Logistics and Operations team and CEO office

Freelance Sept. 2022 - Dec. 2023

Mathematics Tutor

- Taught algebraic concepts to high school students and synthesized 10 reviewers and session materials

KonsultaMD Oct. 2022 - Jan. 2023

Product Intern

- Produced graphs on 7,000 stores across the top 5 pharmacies in 13 regions for competitor analysis
- Generated price comparison data points for 750 medicine products across 4 online pharmacies
- Designed customer journey maps and traits of 4 user personas toward UX and market research

ORGANIZATIONAL EXPERIENCE

DLSU Center for Language Technologies

Member Sept. 2024 - present

- Taught a class for high school students on Machine Learning and Sentiment Analysis

Professor Widom’s Instructional Odyssey at DLSU 2024

Teaching Assistant Sept. 2024

- Assisted the Dean of Stanford University’s School of Engineering along with 4 co-teaching assistants
- Taught 300+ attendees on Spreadsheets, Tableau, Python, and SQL for data science and machine learning

DLSU Association for Computing Machinery Chapter (ACM)

Chair, AY ’23-’24 Sept. 2023 - Aug. 2024

- Supervised a team of 21 officers in the planning and execution of 15 applied computing activities
- Coordinated with international ACM Distinguished Speakers for seminars garnering 100 to 250 attendees

Founding Vice Chair, AY ’22-’23 Sept. 2022 - Aug. 2023

- Chartered a local ACM chapter with 70 members to foster research interest in applied computing fields
- Executed the Open Lab Day to showcase the college’s 10 research labs, drawing over 300 attendees

Peer Tutors Society

Executive Vice President Sept. 2023 - Aug. 2024

- Assisted the president in the management of 12 organization events and 4 committees

Tutor Nov. 2021 - Aug. 2024

- Taught 20+ peers on subjects in discrete structures, programming, artificial intelligence, and mathematics

Vice President for Internal Affairs Sept. 2022 - Aug. 2023

- Supervised website improvements to increase tutor-tutee matching rate from 94.61% to 98.51%
- Matched 335 tutor requests among an average of 30 tutors across 3 terms of an academic year

OTHER AFFILIATIONS

- *Member* - DLSU SPRINT (College of Computer Studies Honors Society)
- *Former Student Member* - Association for Computing Machinery (ACM)
- *Former Research and Academics Chairperson* - CATCH2T25 (College of Computer Studies Batch Government)
- *Former Associate Vice President for Academics* - La Salle Computer Society
- *Former Research and Development Committee Member* - DLSU Microsoft Student Community